

SHEKH EHSANUR RAHMAN

Senior Mobile & Full-Stack AI Engineer • Flutter • React Native • iOS • Android • macOS

Offline-First Architecture • Data Synchronisation • On-Device & LLM AI Features • Founder, Hourwise Labs

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[App Store](#) | [Google Play](#) | [pub.dev packages](#) | [Hourwise](#)

Rajshahi, Bangladesh (UTC+6) • Remote • Three years of shifted hours with a Brazil-based team; available for US/UK/EU overlap • Full-time or independent contractor (B2B)

PROFILE

Senior mobile engineer with 11 years shipping production applications across Flutter, React Native and native Android, iOS and macOS, now building full-stack AI products. 50+ apps published to Google Play and the App Store reaching 1,000,000+ users, including platforms sustaining 100,000+ daily active users at a 4.7-star rating. Specialist in offline-first architecture and conflict-free data synchronisation — author of a published CRDT sync library and nine open-source packages on pub.dev under a verified publisher. Founder and sole engineer of Hourwise Labs, a native macOS billing product live on the Mac App Store that I am extending with on-device AI, a Python backend and cross-platform sync. Experienced team lead and sole owner of products from architecture through CI/CD, store release and post-launch maintenance.

TECHNICAL SKILLS

Cross-Platform & Native: Flutter (7 yrs), Dart, React Native (3+ yrs), TypeScript, Swift, SwiftUI, AppKit, Kotlin, Java, Jetpack Compose — iOS, Android and macOS. Platform channels, native module bridging, custom plugins, white-label and multi-flavor builds

AI & Machine Learning: AI-based invoice token capture (Tanto); on-device naive-Bayes classification trained on user corrections (Hourwise); fully on-device voice command recognition (local_voice); computer vision with homography and registration-mark alignment; AI-assisted engineering with Claude Code under code review. *Currently building:* on-device LLM features with Apple Foundation Models, Claude API integration with structured tool-use output, and retrieval with embeddings and pgvector

Architecture & State: Offline-first and local-first data layers, CRDT conflict-free replication, deterministic conflict resolution, background sync, Clean Architecture, MVVM, dependency injection, BLoC, Cubit, Riverpod, Redux, repository and use-case patterns

Backend & Data: Firebase (Firestore, Auth, Cloud Functions, Crashlytics, FCM), Supabase, PostgreSQL, SQLite (Drift, GRDB), REST API design, webhooks, OAuth2/JWT, biometric auth, encrypted local storage, in-app purchase security; Python and FastAPI (Hourwise backend, in progress)

Quality & Delivery: Unit, widget, integration and end-to-end testing, TDD, structured code review, Crashlytics monitoring, performance profiling, CI/CD pipeline design, Fastlane, GitHub Actions, staged rollouts, App Store and Play Console release management

Specialised: StoreKit 2 subscriptions, EventKit, PDFKit, ARKit/ARCore, shader rendering, WCAG accessibility auditing, maps and routing, payment integration (Pix)

Leadership: Team leadership (6 developers), distributed cross-time-zone collaboration, mentoring, code review standards, technical documentation, stakeholder and client communication

PROFESSIONAL EXPERIENCE

Founder & Full-Stack AI Engineer — Hourwise Labs

09/2026 – Present • Remote — Rajshahi, Bangladesh • Self-employed • [Mac App Store](#) • [Product site](#)

- **Product** — designed, built and shipped Hourwise to the Mac App Store: a native macOS menu-bar app that passively records billable work and turns it into invoices for solo and small-firm lawyers. Sole owner of product direction, engineering, release, support and marketing.
- **Native engineering** — SwiftUI and AppKit (~17k lines of Swift); local-first SQLite/GRDB data layer with versioned migrations, backup and recovery; EventKit calendar import; PDFKit invoices with XLSX, CSV, QuickBooks/Xero and LEDES 1998B exporters; StoreKit 2 subscriptions.
- **On-device intelligence** — work attribution combining a rule engine with a naive-Bayes classifier trained on each user's own corrections, with no account, no server and all data kept on the device.
- **AI roadmap (in progress)** — on-device billing narratives with Apple Foundation Models; private meeting capture and summaries; a Python/FastAPI and PostgreSQL backend with CRDT sync across Swift and Dart clients; a Flutter companion app; and a citation-backed “Ask my matters” agent using the Claude API with retrieval and evaluation.

Senior Mobile Developer (Level III) — Developer eXperience Hub (Devxhub)

09/2023 – 09/2026 • Remote — Rajshahi, Bangladesh

- **Cross-platform architecture** — architected the mobile application layer in Flutter and React Native for enterprise FinTech and expense-management products serving 50K+ users, owning data flow, state management and release process end to end.
- **Offline-first engineering** — built a local/Firebase sync architecture that keeps core functionality fully available through network interruptions, with deterministic conflict resolution on reconnect.
- **Ownership** — team lead and sole developer on multiple client and in-house products, from architecture through CI/CD, store release and maintenance. Delivered Tanto, a Brazilian corporate reimbursement platform with AI-based invoice token capture validated against SEFAZ state tax agencies, policy checks and instant Pix payout; and Farenow, a multi-service super app for a US client (Houston, TX).
- **DevOps** — designed and implemented the team CI/CD pipeline with Fastlane and GitHub Actions, substantially shortening release turnaround for a 4-developer team working across multiple time zones.
- **AI-assisted delivery** — introduced Claude Code into the team workflow while keeping code review and test discipline on all merged work.

- **React Native delivery** — shipped client apps in React Native alongside Flutter: component architecture, Redux and Context state, native module bridging to Kotlin and Swift, and store release on both platforms.
- **Open source** — published 9 reusable packages covering CRDT sync, accessibility auditing, on-device voice, canvas rendering and UI tooling, reducing repeated work across client projects.

Senior Mobile Application Developer — Prabartan Information Technology

06/2019 – 08/2023 · Remote — Rajshahi, Bangladesh

- **Scale** — led development of an offline-first, multi-language EdTech platform reaching 100,000+ students across 5 countries, engineered so lessons and progress remain fully available without connectivity.
- **Reliability** — sustained a 4.7-star store rating at 100,000+ daily active users by building the Firebase/SQLite synchronisation layer and resolving sync-conflict edge cases at scale.
- **Leadership** — managed a 6-developer team and modernised the stack, introducing Flutter 3.0, Jetpack Compose and SwiftUI alongside code review standards.

Android Application Developer — Blue Hills Information Technology

07/2015 – 05/2019 · On-site — Rajshahi, Bangladesh · Part-time alongside university through 04/2017, full-time thereafter

- Co-developed consumer Android applications in Java and Kotlin reaching 500,000+ downloads and a top-10 Education category ranking on Google Play; improved launch time and reduced production defects through profiling and the introduction of code review practices.

Guest Trainer, Mobile App Development (Contract) — NACTAR

2016 – 2024 · 2–3 sessions per year · Bogura, Bangladesh

- Designed and delivered mobile development curriculum to 200+ developers, with hands-on project mentorship aligned to Android and cross-platform certification standards.

OPEN SOURCE — PUBLISHED PACKAGES (PUB.DEV, VERIFIED PUBLISHER)

[flutter_crdt_sync_kit](#) — Offline-first, CRDT-based local data layer with automatic conflict-free merge and pluggable Supabase/REST backend adapters.

[local_voice](#) — Privacy-first, fully on-device voice command recognition — no cloud, no API keys, operates without network access.

[flutter_a11y_lens](#) — Live accessibility auditor inspecting the running widget tree and flagging WCAG violations in real time via an on-screen debug overlay.

Also published: flutter_fabric, flutter_liquid_glass_widgets, flutter_scanner_devxhub, flutter_prayer_time_calculator, flutter_zoom_image, flutter_scroll_date_picker · Full list: pub.dev/publishers/ehsanur.com

SELECTED PRODUCTION WORK

Tanto — Enterprise Reimbursement Platform (Brazil) — AI invoice-token capture validated against SEFAZ state tax agencies, policy checks and instant Pix reimbursement. Company closed Sept 2025; listings removed.

Farenow — Multi-Service Super App (US client, Houston TX) [iOS](#) · [Android](#) — ride-hailing with live tracking, on-demand services, food and grocery delivery, virtual consultations and marketplace listings in a single cross-platform app. Team lead and sole developer, architecture through store release.

Peace of Mind (POM) — Estate Planning App [iOS](#) · [Android](#) — cross-platform app with a solo-built native iOS layer: digital will management, biometric-locked vault, life timeline and StoreKit 2 subscriptions. Resolved subscription race conditions and led App Store submission and compliance.

Also led and built solo, architecture through store release: Muslim Times Pro [iOS](#) · [Android](#) | Notivio [iOS](#) · [Android](#) | ProofSell [iOS](#) · [Android](#) | SK Mobile School [iOS](#) · [Android](#) | [eSIM247](#)

SIDE PROJECTS (GITHUB)

[OMR Scanner](#) — fully offline exam answer-sheet scanner using registration-mark alignment and homography rather than OCR, with grid calibration, live camera guide, auto-capture and instant per-roll scoring.

[Jatri — Bangladesh Transit App](#) — journey planner and ticketing app built on real MRT Line 6 and BRTA route data: search, route results, itinerary detail and ticket purchase flow.

EDUCATION & LANGUAGES

B.Sc. in Computer Science & Engineering — Varendra University, Rajshahi, Bangladesh | 2013 – 2017

Bangla — native · English — professional working proficiency